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Comparative risk of new-onset autoimmune disease following COVID-19 mRNA versus DNA vaccination: A propensity-matched cohort study using the epic cosmos database

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Concerns about autoimmune disease following COVID-19 vaccination persist, yet few studies directly compare autoimmune risk between vaccine platforms. We evaluated whether mRNA vaccines differ from DNA-based vaccines in the incidence of new-onset autoimmune conditions. Using the Epic Cosmos database, we conducted a retrospective matched cohort study of adults ≥18 years without prior autoimmune disease who received either the mRNA vaccine (Pfizer-BioNTech, Moderna) or DNA vaccine (Janssen, AstraZeneca) between December 2020-2022. Patients were propensity score overlap weighted on baseline covariates. Cox models estimated hazard ratios for autoimmune disease over ≥2 years of follow-up. Weighted analyses revealed mRNA vaccination was associated with a higher hazard of autoimmune disease overall versus DNA vaccination (HR 1.045; 95% CI 1.027–1.064; p< 0.001). After post-hoc correction, only vitiligo (HR 1.216; 95% CI 1.069–1.384; p=0.048) showed significant difference. E-values indicate moderate risk of nonvisible confounders. A second cohort from the Apollo dataset (Atropos Health Evidence Network) as a validation set for adults ≥30 years found after propensity score matching, the risk of any autoimmune disease was not statistically significant (HR 1.05; 95% CI 1–1.09; p=0.06). However, the risk of autoimmune dermatologic conditions was statistically significantly higher in the RNA vaccine group (HR 1.12; 95% CI 1.03–1.21; p=0.008). The risk of other autoimmune conditions was not statistically significant (HR 1.01; 95% CI 0.96–1.07; p=0.65). These findings suggest a trend towards higher autoimmune disease risk with conflicting effects in specific disease subsets, supporting further investigation.

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Association between sunscreen use and vitamin D: A cross-sectional analysis of NHANES 2015–2018

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Sunscreen prevents skin damage and decreases the risk of skin cancer; however, the effect of sunscreen use on serum vitamin D levels across sociodemographic groups remains unclear. The goal of this study was to evaluate whether sunscreen use is associated with decreased serum total vitamin D levels. We hypothesized that sunscreen use does not affect serum total vitamin D or vitamin D_3 levels when controlling for demographics and sun-protective behaviors. This population-based cross-sectional study used the National Health and Nutrition Examination Survey (NHANES) data from 2015-2016 and 2017-2018 cycles. 3,921 participants living in the U.S. and with available serum vitamin D lab values and self-reported sunscreen use values were included. Participants were excluded if they used vitamin D supplements or had decreased kidney function. Participants were categorized into four groups based on sunscreen use: Consistent, Sometimes, Rarely, and Never (reference group). Serum total vitamin D and D_3 levels were measured and compared across sunscreen use groups. Adjustments were made for demographics and other sun-protective behaviors. Sunscreen users differed significantly by sex, race, ethnicity, and sun-protective behaviors. Consistent and sometimes sunscreen users were not associated with lower vitamin D levels compared to non-users, even after adjusting for time outdoors. Consistent sunscreen users had significantly higher vitamin D levels (+5.26 nmol/L) than those who never use sunscreen. Additionally, race, ethnicity, and sun-protective behaviors modified the relationship between sunscreen use and vitamin D levels. Sunscreen use is not associated with lower serum vitamin D levels, even when accounting for demographics and sun-protective behaviors. Concerns surrounding vitamin D levels should not discourage regular sunscreen use.

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Swipe, scroll, and stretch marks: A cross-sectional study of striae distensae content on TikTok

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Striae Distensae (SD) are dermal scars caused by excessive tension of the skin. Social media platforms, such as TikTok, increasingly advertise products claiming to reverse SD appearance, yet this content has not been systematically evaluated within the scope of dermatologic research. In this study, we conducted a cross-sectional analysis of 143 TikTok videos on SD content, collected between May 22nd and June 21, 2025, using #stretchmarks and #stretchmarkremoval. Videos were categorized by creator type, and engagement was measured using Video Power Index (VPI) (likes+comments/views). Reliability was assessed by two blinded reviewers using modified DISCERN and JAMA scoring. Spearman's correlation was used to evaluate the relationship between engagement and educational quality. Inter-rater reliability was strong for DISCERN scores (Kw = 0.759 p < .001), weak but statistically significant for JAMA scores (Kw = 0.272 p < .001). Consumers created the majority of SD content (50%) with Physicians accounting for the smallest proportion (13%). Consumers demonstrated the highest engagement (4.53 ± 5.92) and private companies demonstrated the lowest (2.34 ± 1.60). Physicians achieved the highest mean DISCERN scores for Reviewer 1 and 2 (2.78 ± 0.81 and 2.61 ± 0.61 respectively). They also received the highest mean JAMA scores for Reviewer 1 and 2 (1.89 ± 0.32 and 1.28 ± 0.46). Private companies received the lowest mean DISCERN scores for Reviewer 1 and 2 (1.00 ± 0.25 and 0.97 ± 0.30), and consumers demonstrated the lowest mean JAMA scores for Reviewer 1 and 2 (1.00 ± 0.00 and 0.91 ± 0.33). A moderate positive correlation was observed between VPI and DISCERN for both reviewers 1 and 2 (ρ = 0.361 and 0.316; p < .001), while correlations between VPI and JAMA were non-significant. The predominance of high-engagement content produced by consumers, despite lower educational quality, highlights the need for improved dermatologic, evidence-based engagement on TikTok.

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Association of oral isotretinoin with oxidative stress and lipid parameters

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Recent studies in rats suggest that oral isotretinoin may be associated with increased oxidative stress; however, data in humans remain limited. This study aimed to evaluate changes in oxidative stress during oral isotretinoin treatment and their relationship with laboratory parameters. Seventy patients with moderate to severe acne vulgaris and seventy healthy controls were included. In patients, AST, ALT, ALP, GGT, LDL, HDL, triglyceride, and total cholesterol levels were assessed at baseline and at the third month of treatment. Oxidative stress was evaluated using native thiol, total thiol, and disulfide levels measured in controls and in patients before and after therapy. Healthy controls had significantly higher native thiol and total thiol levels than patients (p = 0.010 and p = 0.014), while disulfide levels were similar between groups (p = 0.45). In patients, isotretinoin treatment was associated with increased disulfide levels (p = 0.014) and decreased native and total thiol levels (p = 0.001 and p = 0.002). Post-treatment changes in LDL levels were negatively correlated with native thiol (r = −0.351, p = 0.003) and positively correlated with disulfide levels (r = 0.245, p = 0.041). Similarly, changes in triglyceride levels were negatively correlated with native thiol levels (r = −0.339, p = 0.004) and positively correlated with disulfide levels (r = 0.257, p = 0.032). These findings suggest that oral isotretinoin may increase oxidative stress in humans, reflected by altered thiol–disulfide homeostasis. Increased oxidative stress may contribute to isotretinoin-associated lipid elevations, possibly through lipid peroxidation, and targeting oxidative stress warrants further investigation.